

Question	Marking details	Marks Available					
		AO1	AO2	AO3	Total	Maths	Prac
7	Indicative content A1 Partial pressure of oxygen increases percentage saturation also increases A2 (Hb has) high <u>affinity</u> for O₂ at {high pO₂ / lungs} A3 {Associates / loads} with O₂ (in the lungs / high pO₂) A4 (Hb has) low <u>affinity</u> for O₂ {at low pO₂ / tissues} A5 {dissociates / unloads} with O₂ (in the tissues / low pO₂) A6 sigmoid shape / co-operative binding / steep part of the curve – small drop in pO₂ relatively large % drop in saturation / OWTTE <u>Naked mole rat haemoglobin:</u> B1 Lives in a low pO₂ environment B2 due to {poorly ventilated / underground} burrows / large number of individuals respiring B3 Has <u>Hb</u> with a higher <u>affinity</u> for O₂ B4 <u>Hb</u> can become {fully / higher %} saturated with O₂ B5 {At a lower pO₂ / at same pO₂} as human Hb <u>Hummingbird haemoglobin:</u> C1 Smaller birds so have a larger surface area: volume so lose heat at a faster rate / Flight muscles 30% of their body weight / beat their wings up to 200 times per second C2 have a higher rate of respiration C3 this produces more CO₂ / increase pCO₂ C4 Hb with a lower affinity for O₂ C5 so {dissociates / unloads} more O₂ to the tissues / (more) readily releases O₂	3	3	3	9	0	0

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	7-9 marks Indicative content of this level is detailed content from all three sections <i>The candidate constructs an articulate, integrated account, correctly linking relevant points, such as those in the indicative content, which shows sequential reasoning. The answer fully addresses the question with no irrelevant inclusions or significant omissions. The candidate uses scientific conventions and vocabulary appropriately and accurately.</i> 4-6 marks Indicative content of this level is detailed content from two sections or less detail from three <i>The candidate constructs an account correctly linking some relevant points, such as those in the indicative content, showing some reasoning. The answer addresses the question with some omissions. The candidate usually uses scientific conventions and vocabulary appropriately and accurately.</i> 1-3 marks Indicative content of this level is any correct statement from one area of the indicative content <i>The candidate makes some relevant points, such as those in the indicative content, showing limited reasoning. The answer addresses the question with significant omissions. The candidate has limited use of scientific conventions and vocabulary.</i> 0 marks <i>The candidate does not make any attempt or give a relevant answer worthy of credit.</i>						
	Question 7 total	3	3	3	9	0	0